

Criteria Selection

Persona	WCAG criteria	Tools
Ade	2.1.1, 2.2.1, 2.4.3, 2.4.7, 2.5.5	keyboard_navigation, timing_checker, focus_order_validator, focus_visible_validator, target_size_validator
Elias	1.3.5, 1.4.3, 1.4.12, 2.2.2, 2.4.8	autocomplete_validator, Contrast_Checker (AA), text_formatting, animation_detector, multiple_ways_checker
Ian	1.3.1, 2.2.2, 2.4.6, 3.1.4, 3.1.5	heading_structure, animation_detector, readability_analyzer
Lakshmi	1.1.1, 1.3.1, 2.1.1, 2.4.1, 4.1.2	nvda_agent, heading_structure, keyboard_navigation
Sophie	2.2.1, 2.4.8, 3.1.4, 3.3.1, 3.3.2	timing_checker, multiple_ways_checker, readability_analyzer, form_validator
Stefan	1.4.12, 2.2.2, 2.4.5, 2.4.6, 3.1.4	text_formatting, animation_detector, multiple_ways_checker, heading_structure, readability_analyzer

Tier 1: Strong verification

Manual auditor and tool are checking the same observable thing, and the verdict criteria are concrete.

WCAG	Manual procedure	Tool replication
1.1.1 Non-text Content	Inspect each image, decide if alt text is present and descriptive	axe image-alt rule + screen reader announcement check
1.3.1 Info and Relationships	Read source and check semantic structure, landmarks, list markup	axe + heading_structure_agent

1.3.5 Identify Input Purpose	Inspect form fields for autocomplete attribute	autocomplete_validator (static analysis portion)
1.4.3 Contrast Minimum	Use a contrast picker on text and background	axe color-contrast rule
1.4.12 Text Spacing	Apply WCAG spacing via bookmarklet, check for clipping	text_formatting_agent (your CSS injection approach is exactly this)
2.1.1 Keyboard	Tab through the page, try to operate all controls	keyboard_navigation_agent (needs upgrading from element enumeration to full tab-sequence walk)
2.2.1 Timing Adjustable	Look for timeout messages, try to extend	timing_checker_agent
2.2.2 Pause, Stop, Hide	Look for animations longer than 5 seconds, check for pause controls	animation_detector_agent (the commented Playwright version)
2.4.1 Bypass Blocks	Tab from page start, verify first focusable is a skip link to main	axe skip-link/bypass rules + landmark check
2.4.3 Focus Order	Tab through and compare sequence against visual layout	Needs to be built: capture tab positions, compare to document order
2.4.7 Focus Visible	Tab through and observe focus indicators	Needs to be built: compare computed :focus styles to unfocused state
2.5.5 Target Size	Measure interactive elements with dev tools	target_size_validator_agent
3.1.4 Abbreviations	Scan for abbreviations, check for <code><abbr></code> tags or definitions	readability_analyzer's abbreviation audit
3.3.1 Error Identification	Submit form with errors, check feedback	form_validator's error-handling check
3.3.2 Labels or Instructions	Check form fields for visible labels and instructions	form_validator's label check

4.1.2 Name, Role, Value	Inspect with accessibility tree or screen reader	nvda_agent + axe button-name, aria-required-attr, aria-roles
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Tier 2: Both can verify, but with measurable subjectivity

WCAG	Issue
2.4.8 Location	Manual: easy to check for breadcrumbs and current-location indicators. Tool: multiple_ways_checker_agent does detect breadcrumbs. But "do users know where they are" includes things like page titles, URL structure, and visual cues that involve judgment. Tool checks the structural presence, manual auditor checks the experiential effect.

Tier 3: Both manual and tool struggle

WCAG	Issue
3.1.3 Unusual Words	Manual: requires the auditor to decide what counts as "unusual" for the audience. Two auditors will disagree. Tool: needs a controlled vocabulary or dictionary check, will have false positives on proper nouns and domain terms. Inter-rater agreement is the underlying problem, not the technology.
3.1.5 Reading Level	Manual: WCAG says compare to lower secondary education level, but auditors estimate differently. Tool: produces objective Flesch-Kincaid scores but readability indices don't measure comprehension, only surface text features. A simple sentence about derivatives is still hard.
1.4.8 Visual Presentation (AAA)	Has five sub-requirements (foreground/background customization, line width, justification, line spacing, no horizontal scroll). Some are testable like 1.4.12, others require user preference settings that pages can't fully control. Manual auditor and tool will disagree on what counts.

Final List

Persona	Tier 1 criteria	Tier 3 dropped
Ade	6 (2.1.1, 2.2.1, 2.4.1, 2.4.7, 2.4.3, 2.5.5)	none
Elias	4 (1.3.5, 1.4.12, 1.4.3, 2.2.2) + 2.4.8 in Tier 2	none
Ian	3 (1.3.1, 2.2.2, 3.1.4)	3.1.3, 3.1.5
Lakshmi	5 (1.1.1, 2.1.1, 2.4.1, 4.1.2, 1.3.1)	none
Sophie	3 (3.3.2, 2.2.1, 3.3.1) + 2.4.8 in Tier 2	3.1.5
Stefan	2 (1.4.12, 2.2.2, 3.1.4)	1.4.8, 3.1.3

Dataset Creation

From WCAG ACT Test Suite

[ACT Test Cases | | WAI | W3C](#)

ade 2.1.1 passed=2 | failed=2 | inapplicable=2 [OK]
ade 2.2.1 passed=2 | failed=2 | inapplicable=2 [OK]
ade 2.4.3 passed=0 | failed=0 | inapplicable=0 [PARTIAL]
ade 2.4.7 passed=2 | failed=1 | inapplicable=2 [PARTIAL]
ade 2.5.5 passed=0 | failed=0 | inapplicable=0 [PARTIAL]

elias 1.3.5 passed=2 | failed=2 | inapplicable=2 [OK]
elias 1.4.12 passed=2 | failed=2 | inapplicable=2 [OK]
elias 1.4.3 passed=2 | failed=2 | inapplicable=2 [OK]
elias 2.2.2 passed=2 | failed=1 | inapplicable=2 [PARTIAL]
elias 2.4.8 passed=0 | failed=0 | inapplicable=0 [PARTIAL]

ian 1.3.1 passed=2 | failed=2 | inapplicable=2 [OK]
ian 2.2.2 passed=2 | failed=1 | inapplicable=2 [PARTIAL]
ian 2.4.6 passed=2 | failed=2 | inapplicable=2 [OK]
ian 3.1.4 passed=0 | failed=0 | inapplicable=0 [PARTIAL]
ian 3.1.5 passed=0 | failed=0 | inapplicable=0 [PARTIAL]

lakshmi 1.1.1 passed=2 | failed=2 | inapplicable=2 [OK]
lakshmi 1.3.1 passed=2 | failed=2 | inapplicable=2 [OK]
lakshmi 2.1.1 passed=2 | failed=2 | inapplicable=2 [OK]
lakshmi 2.4.1 passed=2 | failed=1 | inapplicable=1 [PARTIAL]
lakshmi 4.1.2 passed=2 | failed=2 | inapplicable=2 [OK]

sophie 2.2.1 passed=2 | failed=2 | inapplicable=2 [OK]
sophie 2.4.8 passed=0 | failed=0 | inapplicable=0 [PARTIAL]
sophie 3.1.4 passed=0 | failed=0 | inapplicable=0 [PARTIAL]
sophie 3.3.1 passed=2 | failed=2 | inapplicable=1 [PARTIAL]
sophie 3.3.2 passed=0 | failed=0 | inapplicable=0 [PARTIAL]

stefan 1.4.12 passed=2 | failed=2 | inapplicable=2 [OK]
stefan 2.2.2 passed=2 | failed=1 | inapplicable=2 [PARTIAL]
stefan 2.4.5 passed=0 | failed=0 | inapplicable=0 [PARTIAL]
stefan 2.4.6 passed=2 | failed=2 | inapplicable=2 [OK]
stefan 3.1.4 passed=0 | failed=0 | inapplicable=0 [PARTIAL]

From Axe-Core Fixtures

<https://raw.githubusercontent.com/dequelabs/axe-core/master/doc/rule-descriptions.md>

<https://github.com/dequelabs/axe-core/tree/develop/test/integration/rules>

2.2.2 elias, ian, stephan failed=1

3.3.2 sophia passed=2

Hand authored

Per-criterion case counts (unique cases, before persona duplication):

- 2.4.3 passed=2 failed=2 inapplicable=2 (personas: ade)
- 2.4.7 passed=0 failed=1 inapplicable=0 (personas: ade)
- 2.5.5 passed=2 failed=2 inapplicable=2 (personas: ade)

- 2.4.8 passed=2 failed=2 inapplicable=2 (personas: elias, sophie)
- 2.4.1 passed=0 failed=1 inapplicable=1 (personas: lakshmi)

- 3.1.4 passed=2 failed=2 inapplicable=2 (personas: ian, sophie)
- 3.1.5 passed=2 failed=2 inapplicable=2 (personas: ian)

- 2.4.5 passed=2 failed=2 inapplicable=2 (personas: stefan)
- 3.1.4 passed=2 failed=2 inapplicable=2 (personas: stefan)

- 3.3.1 passed=0 failed=0 inapplicable=1 (personas: sophie)
- 3.3.2 passed=2 failed=0 inapplicable=2 (personas: sophie)